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Subject Name: Database Management System

Week 4

Module 2 : Relational Model

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Lecture No: 10:
DDL,DML,TCL,DCL



- A **DBMS** must provide appropriate languages and interfaces for each category of users to express database queries and updates.
- **Database Languages:**
  - communicates with the database
  - used to create and maintain database on computer.
- Mainly two **types** of Database Languages :
  - Data Definition Language (DDL): to specify database schema
  - Data Manipulation Language (DML): to express database queries and updates
- In practical, DDL and DML are not separate languages, instead they are the parts of a single database language such as widely used SQL (Structured Query Language)

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DDL

- **DDL** (stands for **D**ata **D**efinition **L**anguage) is used for specifying the **database schema.**
- Used by the DBA and database designers to specify the conceptual schema of a database
- DDL is used for creating tables, schema, indexes, constraints etc. in database.

  Eg: **create table** *account* (
  account-number **char**(10),
  balance **integer**)

- DDL is used to store the information of **metadata** like the number of tables and schemas, their names, indexes, columns in each table, constraints, etc.
  - DDL compiler generates a set of table templates stored in a *data dictionary*.
  - *Data dictionary contains Metadata (i.e. data about data)*
    - *Database Schema*
      - *Table name*
      - *Column names & types*
    - *Integrity Constraints*
      - *Primary Key*
    - *Authorization*
      - *Who can access what*

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DDL COMMANDS IN SQL

- **CREATE:** used to create the database instance
- **ALTER:** used to alter the structure of the database
- **DROP:** used to delete the database instance
- **TRUNCATE:** used to remove all records from a table
- **RENAME:** used to rename database instances
- **Example**:

```
create table department
    (dept_name char(20),
    building char(15),
    budget numeric(12,2));
```

Note: These commands **either defines or update the database schema** that's why they come under Data Definition language.

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DML

- **DML** (stands for **D**ata **M**anipulation **L**anguage) is used for accessing and manipulating data in a database.
- It allows users to insert, update, delete and retrieve data from the database.
- **DML** is also known as **Query Language**
  - A **query** is a statement requesting the retrieval of information
- Two classes of DML languages:
  - Procedural DMLs/High-level DMLs:
    - User specifies **what** data is required and **how** to get those data
    - Procedural DML is embedded into a high-level programming language like java, etc.
    - Eg: PL/SQL
  - Declarative (Non-Procedural)DMLs/Low-level DMLs:
    - User specifies **what** data is required only; **without** specifying **how** to get those data
    - Declarative DMLs are usually easier to learn and use than procedural DMLs. However, since a user does not have to specify how to get the data, the database system has to figure out an efficient means of accessing data.
    - Eg: SQL



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- **SELECT:** used to retrieve data from a database.
- **INSERT:** used to insert data into a table.
- **UPDATE:** used to update existing data within a table.
- **DELETE:** used to delete all records from a table.

- **Example:**

```
select student_name, Student_age
from student;
```



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DCL

- **Data Control Language (DCL)** is used to control privilege in Databases i.e. DCL is used for granting and revoking user access on a database (Authorization)
- To perform any operation in the database, such as for creating tables, sequences, or views, we need privileges.
- In particular, it is a component of Structured Query Language (SQL)
- **DCL Commands**:
  - **GRANT** – To give user access privileges to a database
  - **REVOKE** – To take back permissions from the user
- The operations for which privileges may be granted to or revoked from a user or role apply to both the Data definition language (DDL) and the Data manipulation language (DML),

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TCL

- **TCL (Transaction Control Language)** commands are used to manage transactions in the database.
- TCL is used to run the changes made by the DML statement.
- The changes in the database that we made using DML commands are either performed or rollbacked using TCL.
- **TCL Commands:**
  - **COMMIT**: to save the transaction on the database
  - **ROLLBACK**: to restore the database to original since the last Commit.
  - **SAVEPOINT**: Savepoint command is used to temporarily save a transaction so that you can rollback to that point whenever necessary.



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SQL SEQUENCE FOR CLAUSE

SELECT ……..

FROM ……..

WHERE ……..

GROUP BY ……..

HAVING ……..

ORDER BY ……..



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Lecture No: 11: SQL: ORDER BY, WHERE (AND, OR, NOT)



ORDER BY CLAUSE

Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
4	Chirag	25	Mumbai	6500.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

- To display details of Emp table in ascending order by *salary*

```
SELECT *
FROM Emp
ORDER BY Salary;
```

BY DEFAULT IN ASC ORDER

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ORDER BY CLAUSE

- OUTPUT:

Emp_ID	Name	Age	Address	Salary
2	Kamal	25	Pune	1500
1	Rahul	32	Delhi	2000
3	Karan	23	Pune	2000
6	Kajal	22	Bilaspur	4500
4	Chirag	25	Mumbai	6500
5	Harsh	32	Mumbai	8500
7	Mahi	24	Patna	10000



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Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
4	Chirag	25	Mumbai	6500.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

- **To sort details of Emp table in ascending order by *name* and *salary***

```
SELECT *
FROM Emp
ORDER BY Name, Salary;
```

In ORDER BY Clause multiple columns can be used.

In this case:

1st column is used as **primary ordering** field,

2nd column is used as **secondary ordering** field,

and so on

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NO SAME NAMES ARE THERE SO SALARY SECONDARY ORDERING WILL NOT BE SEEN IN THIS TABLE

- **OUTPUT:**

Emp_ID	Name	Age	Address	Salary
4	Chirag	25	Mumbai	6500.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
7	Mahi	24	Patna	10000.00
1	Rahul	32	Delhi	2000.00

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Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
4	Chirag	25	Mumbai	6500.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

- **To sort details of Emp table in descending order by *name***

```
SELECT *
FROM Emp
ORDER BY Name DESC;
```


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- **OUTPUT**

Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000
7	Mahi	24	Patna	10000
3	Karan	23	Pune	2000
2	Kamal	25	Pune	1500
6	Kajal	22	Bilaspur	4500
5	Harsh	32	Mumbai	8500
4	Chirag	25	Mumbai	6500



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SQL: WHERE_____AND , OR, NOT OPERATORS

Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
4	Chirag	25	Mumbai	6500.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

Selects all records from *Emp* table whose *salary* is greater than 2000 and age is less than 25

```
SELECT *
FROM Emp
WHERE Salary>2000 AND Age<25;
```



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- **OUTPUT:**

Emp_ID	Name	Age	Address	Salary
6	Kajal	22	Bilaspur	4500
7	Mahi	24	Patna	10000

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Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
4	Chirag	25	Mumbai	6500.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

Selects all records from *Emp* table whose *salary* is greater than 2000 or whose age is less than 25

```
SELECT *
FROM Emp
WHERE Salary>2000 OR Age<25;
```



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- **OUTPUT**

THIS SATISFIES THE AGE CONDITION BUT NOT THE SALARY CONDITION

Emp_ID	Name	Age	Address	Salary
3	Karan	23	Pune	2000
4	Chirag	25	Mumbai	6500
5	Harsh	32	Mumbai	8500
6	Kajal	22	Bilaspur	4500
7	Mahi	24	Patna	10000

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Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
4	Chirag	25	Mumbai	6500.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

Selects all records where address is *not* Mumbai

```
SELECT *
FROM Emp
WHERE NOT Address='Mumbai';
```



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- **OUTPUT:**

Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.0
2	Kamal	25	Pune	1500.0
3	Karan	23	Pune	2000.0
6	Kajal	22	Bilaspur	4500.0
7	Mahi	24	Patna	10000.

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COMBINING NOT AND OR

Emp_ID	Name	Age	Address	Salary
1	Rahul	32	Delhi	2000.00
2	Kamal	25	Pune	1500.00
3	Karan	23	Pune	2000.00
4	Chirag	25	Mumbai	6500.00
5	Harsh	32	Mumbai	8500.00
6	Kajal	22	Bilaspur	4500.00
7	Mahi	24	Patna	10000.00

Selects all records from *Emp* table whose *salary* is greater than 2000 and age is less than 25 or whose Emp ID is 2.

```
SELECT *
FROM Emp
WHERE (Salary>2000 AND Age<25) OR (Emp_ID=2);
```


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- **OUTPUT:**

Emp_ID	Name	Age	Address	Salary
2	Kamal	25	Pune	1500.0
6	Kajal	22	Bilaspur	4500.0
7	Mahi	24	Patna	10000.

SATISFY THE AND CONDITION

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Lecture No: 12: GROUPBY + AGGREGATE FUNCTIONS(MIN,MAX,COUNT,SUM,AVG)



SQL: GROUP BY CLAUSE WITH AGGREGATE FUNCTIONS (min, max, count,avg)

Emp_ID	Name	Age	Address	Salary	Dept_ID
1	Rahul	32	Delhi	2000.00	500
2	Kamal	25	Pune	1500.00	500
3	Karan	23	Pune	2000.00	500
4	Chirag	25	Mumbai	6500.00	501
5	Harsh	32	Mumbai	8500.00	501
6	Kajal	22	Bilaspur	4500.00	502
7	Mahi	24	Patna	10000.00	502

Write a query to find the number of employee in each department

```
SELECT Dept_id, COUNT(*)
FROM Emp
GROUP BY Dept_id
```

COUNTS TOTAL NUMBER OF RECORDS IN EACH GROUP

SHOULD BE SAME

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- **OUTPUT:**

Dept_ID	COUNT(*)
500	3
501	2
502	2

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Emp_ID	Name	Age	Address	Salary	Dept_ID
1	Rahul	32	Delhi	2000.00	500
2	Kamal	25	Pune	1500.00	500
3	Karan	23	Pune	2000.00	500
4	Chirag	25	Mumbai	6500.00	501
5	Harsh	32	Mumbai	8500.00	501
6	Kajal	22	Bilaspur	4500.00	502
7	Mahi	24	Patna	10000.00	502

Write a query to find the number of employee in each department, sorted low to high

```
SELECT Dept_id, COUNT(*)
FROM Emp
GROUP BY Dept_id
ORDER BY COUNT(*);
```



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- **OUTPUT:**

Dept_ID	COUNT(*)
501	2
502	2
500	3

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Emp_ID	Name	Age	Address	Salary	Dept_ID
1	Rahul	32	Delhi	2000.00	500
2	Kamal	25	Pune	1500.00	500
3	Karan	23	Pune	2000.00	500
4	Chirag	25	Mumbai	6500.00	501
5	Harsh	32	Mumbai	8500.00	501
6	Kajal	22	Bilaspur	4500.00	502
7	Mahi	24	Patna	10000.00	502

Write a query to find the sum of salaries of employees from each department

```
SELECT Dept_id, SUM(Salary) AS 'SumSalary'
FROM Emp
GROUP BY Dept_id;
```

AS USE FOR RENAME

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- **OUTPUT:**

Dept_ID	SumSalary
500	5500.00
501	15000.00
502	14500.00

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Emp_ID	Name	Age	Address	Salary	Dept_ID
1	Rahul	32	Delhi	2000.00	500
2	Kamal	25	Pune	1500.00	500
3	Karan	23	Pune	2000.00	500
4	Chirag	25	Mumbai	6500.00	501
5	Harsh	32	Mumbai	8500.00	501
6	Kajal	22	Bilaspur	4500.00	502
7	Mahi	24	Patna	10000.00	502

Write a query to find the average of salaries of employees from each department

```
SELECT Dept_id, ROUND(AVG(Salary))
FROM Emp
GROUP BY Dept_id;
```

ROUND OFF OPERATION



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- **OUTPUT:**

Dept_ID	AVG(Salary)
500	1833
501	7500
502	7250

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Emp_ID	Name	Age	Address	Salary	Dept_ID
1	Rahul	32	Delhi	2000.00	500
2	Kamal	25	Pune	1500.00	500
3	Karan	23	Pune	2000.00	500
4	Chirag	25	Mumbai	6500.00	501
5	Harsh	32	Mumbai	8500.00	501
6	Kajal	22	Bilaspur	4500.00	502
7	Mahi	24	Patna	10000.00	502

Write a query to find the minimum salary of employee from each department

```
SELECT Dept_id, MIN(Salary)
FROM Emp
GROUP BY Dept_id;
```



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- **OUTPUT:**

Dept_ID	MIN(Salary)
500	1500.00
501	6500.00
502	4500.00

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Emp_ID	Name	Age	Address	Salary	Dept_ID
1	Rahul	32	Delhi	2000.00	500
2	Kamal	25	Pune	1500.00	500
3	Karan	23	Pune	2000.00	500
4	Chirag	25	Mumbai	6500.00	501
5	Harsh	32	Mumbai	8500.00	501
6	Kajal	22	Bilaspur	4500.00	502
7	Mahi	24	Patna	10000.00	502

Write a query to find the maximum salary of employee from each department

```
SELECT Dept_id, MAX(Salary)
FROM Emp
GROUP BY Dept_id;
```



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- **OUTPUT:**

Dept_ID	MAX(Salary)
500	2000.00
501	8500.00
502	10000.00

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Emp_ID	Name	Age	Address	Salary	Dept_ID
1	Rahul	32	Delhi	2000.00	500
2	Kamal	25	Pune	1500.00	500
3	Karan	23	Pune	2000.00	500
4	Chirag	25	Mumbai	6500.00	501
5	Harsh	32	Mumbai	8500.00	501
6	Kajal	22	Bilaspur	4500.00	502
7	Mahi	24	Patna	10000.00	502

Write a query to find the minimum and maximum salary of employee from each department

```
SELECT Dept_id, MIN(Salary), MAX(Salary)
FROM Emp
GROUP BY Dept_id;
```



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- **OUTPUT:**

Dept_ID	MIN(Salary)	MAX(Salary)
500	1500.00	2000.00
501	6500.00	8500.00
502	4500.00	10000.00

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Emp_ID	Name	Age	Address	Salary	Dept_ID
1	Rahul	32	Delhi	2000.00	500
2	Kamal	25	Pune	1500.00	500
3	Karan	23	Pune	2000.00	500
4	Chirag	25	Mumbai	6500.00	501
5	Harsh	32	Mumbai	8500.00	501
6	Kajal	22	Bilaspur	4500.00	502
7	Mahi	24	Patna	10000.00	502

Write a query to find the number of employees in each department location wise

```
SELECT Dept_id, Address, COUNT(*)
FROM Emp
GROUP BY Dept_id, Address;
```



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Emp_ID	Name	Age	Address	Salary	Dept_ID
1	Rahul	32	Delhi	2000.00	500
2	Kamal	25	Pune	1500.00	500
3	Karan	23	Pune	2000.00	500
4	Chirag	25	Mumbai	6500.00	501
5	Harsh	32	Mumbai	8500.00	501
6	Kajal	22	Bilaspur	4500.00	502
7	Mahi	24	Patna	10000.00	502

GROUP BY Dept_ID

Group by address/location

OUTPUT:

Dept_ID	Address	COUNT(*)
500	Delhi	1
500	Pune	2
501	Mumbai	2
502	Bilaspur	1
502	Patna	1

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Thank You



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